shared index is the process of tensor contraction, since we start with, for example, two input tensors with two indices each (for a total of four indices) and the output has two indices because two of the four get contracted away. If we did a tensor contraction over two 3-tensors, the result would be a 4-tensor.

Tensor contraction: example

Let's tackle a concrete example of a tensor contraction; we'll contract two matrices using Einstein notation.

$$A = \begin{bmatrix} 1 & -2 & 4 \\ -5 & 9 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} -3 & -3 \\ 5 & 9 \\ 0 & 7 \end{bmatrix}$$

Matrix A is a 2 × 3 matrix and matrix B is 3 × 2. We will label the dimensions of these matrices using arbitrary characters. For example, we'll label matrix A:$i \times j$ with dimensions (indices) i and j, and matrix B:$j \times k$ with indices j and k. We could have labeled the indices using any characters, but we want to contract over the shared dimensions of $A_j = B_j = 3$, so we label them with the same characters.

$$C = \begin{bmatrix} x_{0,0} & x_{0,1} \\ x_{1,0} & x_{1,1} \end{bmatrix}$$

This C matrix represents the output. Our goal is to figure out the values of the x values, which are labeled by their indexed positions. Using the previous tensor contraction formula, we can figure out $x_{0,0}$ by finding row 0 of matrix A and column 0 of matrix B:$A_{0,j} = [1,–2,4]$ and $B_{j,0} = [–3,5,0]^T$. Now we loop over the j index, multiplying each element of $A_{0,j}$ with $B_{j,0}$ and then summing them together to get a single number, which will be $x_{0,0}$ In this case, $x_{0,0} = \Sigma A_{0,j} \times B_{j,0} = (1 \times –3) + (–2 \times 5) + (4 \times 0) = –3 – 10 = –$